SkillUp

Latest Trends and Tools

Objectives

After completing this reading, you will be able to:

- Explore the latest trends in generative AI for BI
- Explore cutting-edge tools with practical applications of AI technologies

Introduction

Generative AI is revolutionizing business intelligence (BI) by transforming data generation, analysis, and visualization. This technology enables organizations to gain actionable insights, enhance strategic decision-making, and stay competitive in the marketplace.

Key Trends Transforming Business Intelligence with AI

Augmented Analytics

Augmented analytics uses AI and machine learning to improve data preparation, insight generation, and visualization. It makes analytics more accessible and powerful by automating complex processes and providing intelligent recommendations.

Continuous Intelligence Continuous intelligence processes real-time data to support instant decision-making. Integrating AI with streaming data platforms provides timely insights, enabling quick responses to emerging trends and anomalies.

Explainable AI (XAI) Explainable AI aims to make AI models transparent and understandable. BI builds trust and accountability by clearly explaining how AI-generated insights and predictions are derived.

AI-Driven Data Governance AI-driven data governance automates the management of data quality, privacy, and compliance. This ensures that BI processes use accurate, secure, and regulation-compliant data, reducing the risk of breaches and penalties.

Federated Learning Federated learning trains AI models across decentralized data sources without sharing sensitive information. This enhances privacy and security while leveraging diverse data sets for more robust and accurate BI insights.

Innovative AI Tools Transforming Business Intelligence

H2O.ai (H2O.ai | Convergence of The World' s Best Predictive & Generative AI)

H2O.ai offers an open-source platform for building and deploying AI models. It features automated machine learning (AutoML) capabilities that simplify the development and deployment of advanced analytics solutions for BI professionals.

ThoughtSpot (https://www.thoughtspot.com/)

ThoughtSpot is a search and AI-driven analytics platform that enables users to perform natural language queries on their data. Its AI capabilities provide instant insights, making data exploration intuitive and accessible for non-technical users.

KNIME (https://www.knime.com/)

KNIME is an open-source data analytics platform integrating machine learning and data processing tools. It supports various data sources and offers an intuitive interface for building and deploying data workflows.

Domo (https://www.domo.com/)

Domo is a cloud-based BI platform that combines data integration, visualization, and AI-powered analytics. It provides real-time data access and advanced analytics features, enabling organizations to make data-driven decisions quickly.

BigML (https://bigml.com/)

BigML provides an easy-to-use platform for creating and deploying machine learning models. It emphasizes simplicity and accessibility, making advanced analytics capabilities available to a broader audience.

Summary:

Generative AI is transforming BI with innovative trends and powerful tools.

Key trends driving BI advancements are:

- **Augmented analytics:** Uses AI and machine learning to enhance data analysis and insights.
- **Continuous intelligence:** Enables real-time decision-making through continuous data processing.
- **Explainable AI:** Improves transparency and understanding of AI models.
- **AI-driven data governance:** Automates data quality, privacy, and compliance management.
- **Federated learning:** Trains AI models across decentralized data sources to enhance privacy and security.

Tools demonstrating these trends:

- **H2O.ai:** Open-source platform for building and deploying AI models.
- **ThoughtSpot:** AI-driven analytics platform for natural language data queries.
- **KNIME:** Open-source data analytics platform with machine learning integration.
- **Domo:** Cloud-based BI platform combining data integration, visualization, and AI-powered analytics.
- **BigML:** User-friendly platform for creating and deploying machine learning models.

These trends and tools help organizations leverage AI for superior business outcomes.